

Capacity Management (Answer Key)

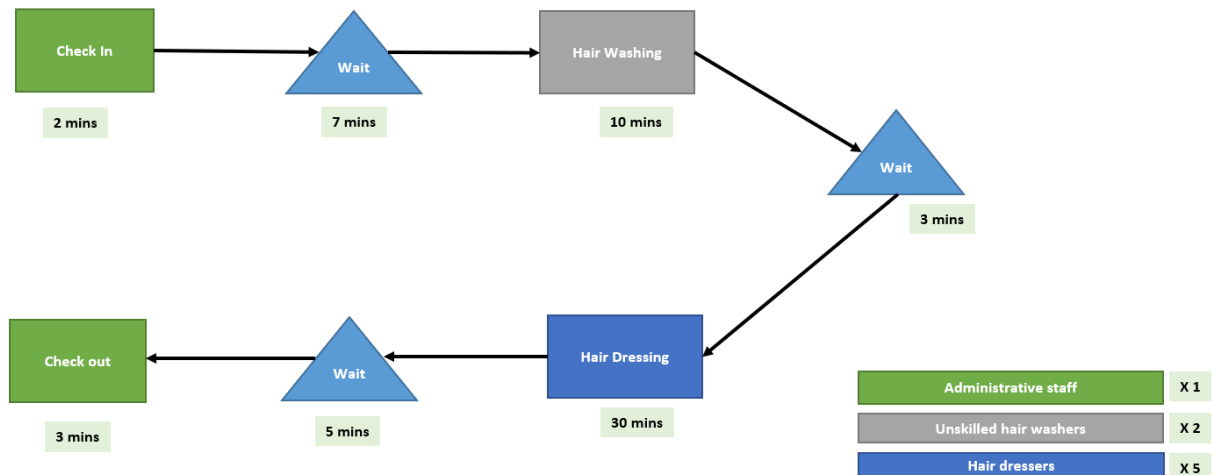
Task 1:

Consider the hairdresser firm ProfiCutZ. Customers of this hairdresser proceed according to the following steps. They enter the shop, immediately check in (which takes 2 minutes per customer). Then customers wait 7 minutes. Then they proceed to hair-washing, which takes 10 minutes per customer. Then they wait again for 3 minutes. Next, their hair is being cut by a professional hairdresser (this step is called hairdressing), which takes 30 minutes. After waiting another 5 minutes, customers proceed to check-out to pay, and leave the shop. This entire check-out procedure takes another 3 minutes.

Assume there are three groups of workers in the firm ProfiCutZ: professional hairdressers, hair washers, and administration. Professional hairdressers can perform any task, but usually only engage in hairdressing. Hair washers can only wash hair. Administration staff can only accommodate the check-in and check-out procedures. Currently, ProfiCutZ employs 5 hairdressers, 2 unskilled hair washers, and 1 administrative person.

- a. Perform a process flow chart analysis for the firm ProfiCutZ: Draw a process flow diagram.

Answer:



- b. What is the capacity of ProfiCutZ in customers per hour?

Answer: 10 customers/hour

- c. Which resource is the bottleneck at ProfiCutZ?

Answer: Hair dressing

- d. How many customers are there, at the same time, in 1 hour at ProfiCutZ in the steady state?

Answer: 10 customers

- e. From the identified bottleneck, the company decides to hire 2 employees who can perform these tasks. What is the maximum possible capacity at ProfiCutZ with these 2 employees?

Answer: 13 customers/hour.

Task 2:

Circored plant converts iron ore (in the form of iron ore fines) into direct reduced iron (DRI) briquettes. The plant has an annual demand of 657,000 tons of DRI briquettes. The iron ore is the flow unit in the process as it undergoes multiple transformation before becoming DRI briquettes.

In the Circored plant, the first step the flow unit encounters in the process is the preheater, where the iron ore fines (which have a texture like large-grained sand) are dried and heated. The heating is achieved through an inflow of high-pressured air, which is blown into the preheater from the bottom. The high-speed air flow "fluidizes" the ore, meaning that the mixed air-ore mass (a "sandstorm") circulates through the system as if it was a fluid, while being heated to a temperature of approximately 850-900°C. The preheater can process 120 tons per hour.

From the preheater, a large bucket elevator transports the ore to the second process step, the lock hoppers. The lock hoppers consist of three large containers, separated by sets of double isolation valves. Their role is to allow the ore to transition from an oxygen-rich environment to a hydrogen atmosphere. The lock hoppers can process 110 tons per hour.

Following the lock hoppers, the ore enters the circulating fluid bed reactor (CFB), where the actual reduction process begins. The reduction process requires the ore to be in the reactor for 15 minutes. At any given moment of time, there can be, at maximum, 28 tons in the CFB reactor.

After this first reduction, the material flows into the stationary fluid bed reactor (SFB, 2nd reactor). This second reaction takes about four hours. The reactor is the size of a medium two-family home and contains 400 tons of the hot iron ore at any given moment in time. Stationary fluid bed reactors require 4 hours for the material to react.

Following the fluid bed reactors, the reduced iron enters the flash heater, in which a stream of high-velocity hydrogen carries the DRI to the top of the plant while simultaneously reheating it to a temperature of 685 °C. The flash heater can process 135 tons per hour.

After the flash heater, the DRI enters the pressure let-down system (discharger). As the material passes through the discharger, the hydrogen atmosphere is gradually replaced by inert nitrogen gas. Pressure and hydrogen are removed in a reversal of the lock hoppers at the beginning. Hydrogen gas sensors assure that material leaving this step is free of hydrogen gas and, hence, safe for briquetting. The pressure let-down system has a capacity of 118 tons per hour.

Each of the three briquetting machines contains two wheels that turn against each other, each wheel having the negative of one-half of a briquette on its face. The DRI is poured onto the wheels from the top and is pressed into briquettes, or iron bars, which are then moved to a large pile of finished goods inventory. Each of the three briquetting machines has a capacity of 55 tons per hour. These work in parallel.

- a. What is the overall capacity of the Circored plant when the plant is operating in Steady state?

Answer: 100 tons/hour

- b. How long does it take to produce 25,000 tons of DRI iron Briquettes?

Answer: 333.33 hours

- c. What is the average capacity utilization of each resource?

Answer: 75%

Task 3:

An old apartment in Heilbronn needs to be renovated. The apartment owner has contacted "Remodel your Home" a company specializing in home renovation. The project manager has visited the apartment and has prepared the following time schedule for all the necessary activities.

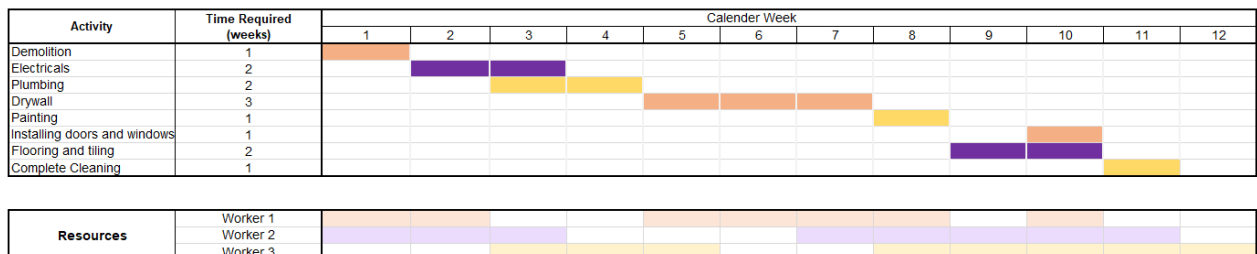
Activity	Predecessor Activity	Time Required (weeks)
Demolition	-	1
Electricals	Demolition	2
Plumbing	Demolition	2
Drywall	Electricals & Plumbing	3
Painting	Drywall	1
Installing doors and windows	Painting	1
Flooring and tiling	Painting	2
Complete Cleaning	Flooring and tiling	1

The renovation will begin from CW1 of 2023, and the project manager has decided to employ three workers for this project. Each worker is capable of performing specific activities and is available during the calendar weeks mentioned below.

Worker	Activity	Availability
Worker 1	Demolition Drywall Installing doors and windows	CW1 - CW2 CW5 - CW8 CW10
Worker 2	Electricals Flooring and tiling Installing doors and windows	CW1 - CW3 CW7 - CW11
Worker 3	Plumbing Painting Complete Cleaning	CW3 - CW5 CW8 - CW12

- a. Draw a Gantt Chart of the renovation project. The project starts as early as possible.

Answer:



- b. How long (in weeks) does it take for the project to complete once the renovation has begun (assuming no public holidays come in between)?

Answer: 11

- c. Is there a week(s) when the renovation work is put on hold? If so, which week(s)?

Answer: No